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Online Conformal Prediction via Universal Portfolio Algorithms

A Parameter-Free Reduction Turns Coverage Tuning into Portfolio Selection

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Online conformal prediction promises long-run coverage even on adversarial data streams—but every prior method needed a hand-tuned learning rate to actually deliver it. This paper shows that controlling a general notion of *linearized regret* implies coverage for *any* online algorithm, then reduces the resulting regret-minimization problem to two-asset portfolio selection. The result, **UP-OCP**, is parameter-free by construction, borrowing decades of tuning-free universal portfolio algorithms to deliver calibrated coverage without a single knob to turn.

AT A GLANCE

Problem: Online conformal methods need a hand-tuned learning rate to guarantee coverage—exactly the kind of choice adversarial streams make impossible to validate in advance.

Why it matters: Mistuned online conformal intervals silently lose their coverage guarantee on the non-stationary or adversarial streams they are meant to handle.

Method: A regret-to-coverage reduction via the Fenchel conjugate, followed by a reduction of the resulting problem to two-asset portfolio selection.

Result: UP-OCP matches or exceeds tuned ACI-style baselines’ coverage with zero learning-rate parameters.

Evidence: Comparisons against Adaptive Conformal Inference (ACI) variants on time-series and adversarial streams.

The Big Picture

Online conformal prediction (OCP) methods maintain a calibrated threshold as data arrive one at a time, aiming for long-run $(1 - \alpha)$ coverage even when the stream is non-stationary or adversarial. The dominant family, descended from Adaptive Conformal Inference (ACI), updates the threshold via an online-gradient-style rule governed by a learning rate. That learning rate trades off how quickly the method tracks distribution shift against how much

it overreacts to noise—and the correct value is stream-dependent and unknowable in advance, precisely in the adversarial setting OCP exists to handle without such foreknowledge.

This paper asks a structural question instead of a tuning question: what property of an online algorithm’s regret actually implies coverage, for any algorithm at all?

Why Existing Approaches Fall Short

ACI-style methods require an algorithm-specific regret analysis for every new update rule, and their coverage guarantees are only as good as the learning rate chosen for that specific rule. Prior refinements made the tuning more forgiving but never removed it. Because the correctness of the guarantee depends on a hyperparameter that cannot be cross-validated on an adversarial stream without assuming away the very adversariality the method is supposed to survive, tuning is not merely inconvenient—it is a genuine gap in the coverage guarantee’s practical reliability.

Core Mechanism

The authors show that controlling *linearized regret* for the $(1 - \alpha)$ -pinball loss implies a long-run coverage bound for any online algorithm, via a black-box reduction that depends only on the Fenchel conjugate of an upper bound on that regret. This decouples “does it cover” from “which algorithm produced the threshold.” Recognizing that the threshold-update problem, viewed through this regret lens, is structurally identical to sequential betting between two assets (covered vs. not covered), they import universal portfolio algorithms—originally designed for sequential investing with no prior knowledge of asset returns—directly into the conformal setting.

Technical Formulation

Let s_t be the nonconformity score at time t and q_t the calibrated threshold, with pinball loss

$$\rho_{1-\alpha}(u) = \begin{cases} (1 - \alpha)u & u \geq 0 \\ -\alpha u & u < 0 \end{cases}, \quad u = q_t - s_t.$$

Define regret

$$R_T = \sum_{t=1}^T \rho_{1-\alpha}(q_t - s_t) - \min_q \sum_{t=1}^T \rho_{1-\alpha}(q - s_t).$$

Empirical coverage error is controlled by R_T through a bound derived from the Fenchel conjugate of the regret’s upper envelope. Casting q_t ’s update as a bet fraction between a “covered” and “not covered” asset reduces the problem to constant-rebalanced portfolio selection, for which universal portfolio algorithms guar-

antee $O(\log T)$ regret with no learning-rate parameter—translating, through the regret-to-coverage bound, into parameter-free coverage.

Learning or Inference Procedure

1. Observe nonconformity score s_t at each round t .
2. Map the threshold-selection problem to a bet fraction over a two-asset (covered / not covered) portfolio.
3. Update the bet using a universal portfolio algorithm—no learning rate is set or tuned.
4. Convert the resulting bet fraction back into the calibrated threshold q_{t+1} and output the prediction interval.

What the Guarantee Says

For any sequence of scores, including adversarially chosen ones, the empirical long-run miscoverage rate deviates from α by an amount controlled by $R_T/T \rightarrow 0$, where R_T is the universal portfolio algorithm’s $O(\log T)$ regret. The bound holds without assuming any distributional structure on the stream and without any user-chosen learning rate.

Experimental Findings

On time-series and deliberately adversarial data streams, UP-OCP matches or exceeds the empirical coverage of ACI variants that required per-stream learning-rate tuning, while UP-OCP itself uses none—removing an entire axis of hyperparameter search from online conformal deployment.

Ablations and Interpretation

Against best-tuned ACI: Even when ACI’s learning rate is tuned in hindsight on each specific stream, UP-OCP’s parameter-free coverage remains competitive, showing little practical cost to removing the tuning step.

Against mistuned ACI: When ACI’s learning rate is chosen without stream-specific tuning, its coverage degrades substantially; UP-OCP’s guarantee is unaffected, since it never depended on a tuned parameter in the first place.

Beyond intervals: The authors note the regret-to-coverage reduction is not intrinsically tied to the two-asset case, suggesting the portfolio-algorithm connection may extend to richer conformal prediction settings.

Reference

Tuo Liu, Edgar Dobriban, Francesco Orabona. **Online Conformal Prediction via Universal Portfolio Algorithms.** arXiv:2602.03168, February 2026.
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